Student Performance Trend Analysis

1. Problem Statement:

Compare student marks across multiple examinations, calculate improvements and declines, identify weak subjects, and classify student performance.

2. Objectives:

- Compare examination marks.
- Calculate improvement or decline.
- Identify weak subjects.
- Classify student performance.

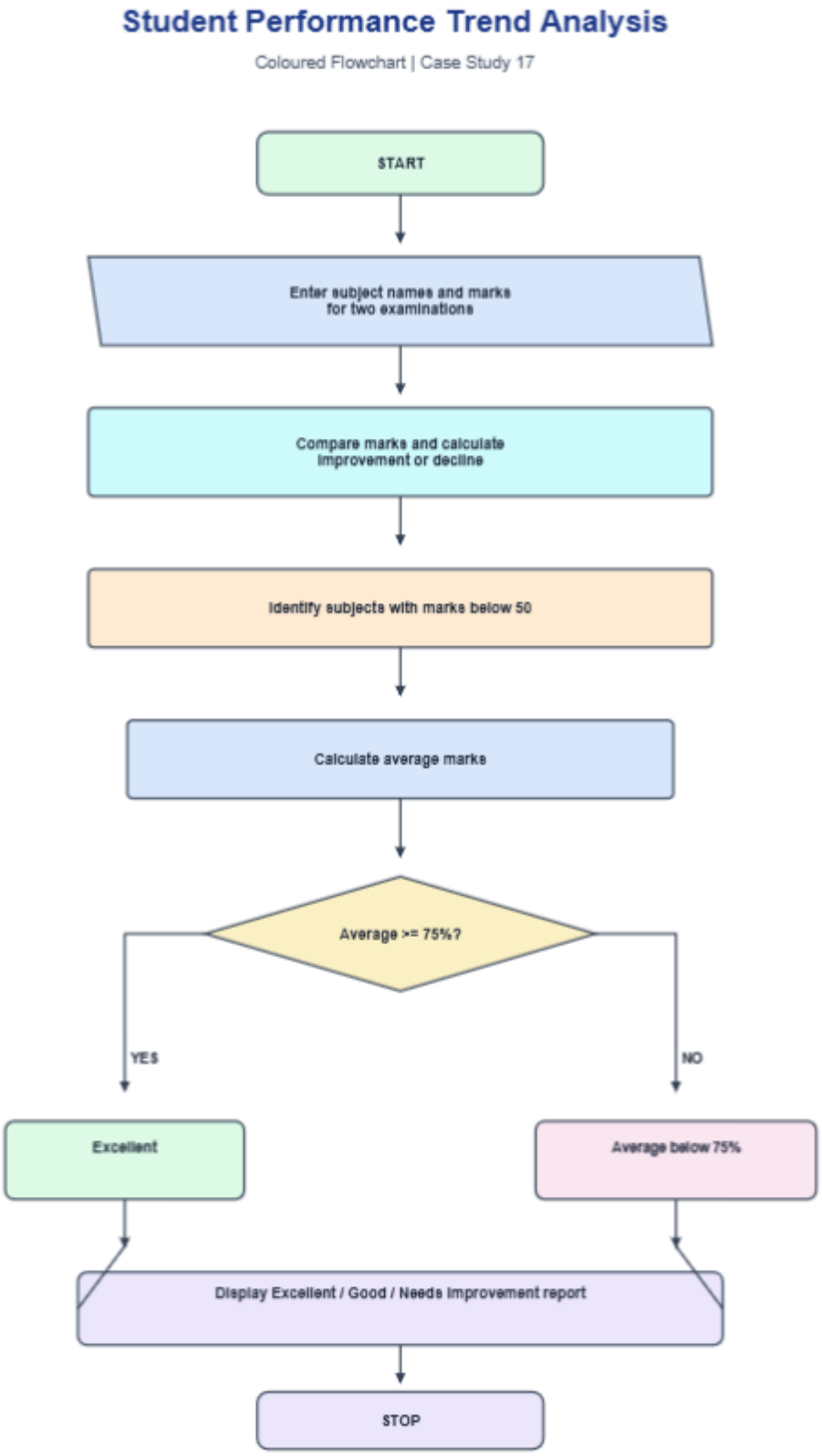
3. Assumptions:

- Marks are between 0 and 100.
- Minimum passing marks are 50.
- Performance is classified based on average marks.

4. Algorithm:

1. Start.
2. Enter subject names and examination marks.
3. Compare marks in two examinations.
4. Calculate improvement or decline.
5. Identify subjects with marks below 50.
6. Calculate average marks.
7. Classify student performance.
8. Display the results.
9. Stop.

5. Flowchart:



6. Pseudocode:

```
None
START
INPUT subject names and exam marks

FOR each subject
    change = exam2 - exam1
    DISPLAY change

    IF exam2 < 50 THEN
        DISPLAY "Weak Subject"
    END IF
END FOR

average = total marks / number of subjects

IF average >= 75 THEN
    DISPLAY "Excellent"
ELSE IF average >= 50 THEN
    DISPLAY "Good"
ELSE
    DISPLAY "Needs Improvement"
END IF

STOP
```

8. Sample Output:

```
None
Python
Improvement/Decline: 15
Good Performance

Maths
Improvement/Decline: -5
Weak Subject

English
Improvement/Decline: 10
Good Performance
```

```
Average Marks: 65.0
Good
```

9. Test Cases:

Subject	Exam 1	Exam 2	Result
Python	60	75	Improved
Maths	45	40	Declined
English	70	80	Improved

10. Conclusion:

This program compares examination marks, identifies improvements and weak subjects, and classifies student performance based on average marks.